



(12) **United States Plant Patent**
van der Meer

(10) **Patent No.:** **US PP22,000 P3**
(45) **Date of Patent:** **Jun. 28, 2011**

(54) **ECHINACEA PLANT NAMED ‘CATHARINA’**

(50) Latin Name: *Echinacea purpurea*
Varietal Denomination: **Catharina**

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(NL)

(*) Notice: Subject to any disclaimer, the term of this
patent is extended or adjusted under 35
U.S.C. 154(b) by 0 days.

(21) Appl. No.: **12/589,198**

(22) Filed: **Oct. 19, 2009**

(65) **Prior Publication Data**

US 2011/0093995 P1 Apr. 21, 2011

(51) **Int. Cl.**
A01H 5/00 (2006.01)

(52) **U.S. Cl.** **Plt./428**

(58) **Field of Classification Search** Plt./428
See application file for complete search history.

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(57) **ABSTRACT**

A new and distinct *Echinacea purpurea* cultivar named
‘CATHARINA’ is disclosed, characterized by distinctive
double dark pink flowers and uniquely strong peduncles. The
new cultivar also have unique ray florets that resist drooping
after harvest. The new variety is an *Echinacea*, and is nor-
mally used as a garden or container plant, and also suitable for
cut flower production.

1 Drawing Sheet

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Latin name of the genus and species: *Echinacea purpurea*.
Variety denomination: ‘CATHARINA’.

BACKGROUND OF THE INVENTION

The new cultivar was discovered as a chance, whole plant
mutation in a commercial nursery in Rijpwetering, the Neth-
erlands. The inventor, Cornelis Petrus van der Meer, a citizen
the Netherlands, discovered the new variety as a naturally
occurring whole plant mutation among several plants of the
unpatented commercial variety, *Echinacea purpurea* ‘Mag-
nus’. The new variety was discovered in Spring of 2006 by the
inventor.

Asexual reproduction of the new cultivar ‘CATHARINA’
by vegetative divisions was performed at a commercial nurs-
ery in Rijpwetering, the Netherlands in the Summer of 2006.
Subsequently, reproduction has also been successful by tissue
culture. Vegetative reproduction by tissue culture has pro-
duced several generations, and has shown that the unique
features of this cultivar are stable and reproduced true to type
through successive generations.

SUMMARY OF THE INVENTION

The cultivar ‘CATHARINA’ has not been observed under
all possible environmental conditions. The phenotype may
vary somewhat with variations in environment such as tem-
perature, day length, and light intensity, without, however,
any variance in genotype.

The following traits have been repeatedly observed and are
determined to be the unique characteristics of ‘CATHA-
RINA.’ These characteristics in combination distinguish
‘CATHARINA’ as a new and distinct *Echinacea purpurea*
cultivar:

1. Distinctive double dark pink flowers.
2. Uniquely strong peduncles.
3. Suitability for garden uses, and commercial cut flower
production.
4. Durability of ray florets to resist drooping after harvest.

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PARENT COMPARISON

Plants of the new cultivar ‘Catharina’ are similar to plants
of the parent variety *Echinacea purpurea* ‘Magnus’ in most
horticultural characteristics. However, plants of the new cul-
tivar ‘CATHARINA’ produce double flowers, or a different
color of pink. Additionally, ray florets of ‘Magnus’ will droop
within a few days of harvesting flowers for cut flower pur-
poses. Cut flowers of ‘Catharina’ will resist drooping ray
florets for a minimum of 7 days.

COMMERCIAL COMPARISON

In addition to the parent variety, ‘Catharina’ can also be
compared to the patented, commercially known variety
Echinacea purpurea ‘Pink Double Delight,’ U.S. Plant Pat.
No. 18,803. The new variety produces overall darker pink
flowers, on a shorter plant. Additionally the ray florets of the
new cultivar have a different shape, droop less and remain
intact longer after harvesting a cut flower.

BRIEF DESCRIPTION OF THE PHOTOGRAPH

The accompanying photograph in FIG. 1 illustrates in full
color a typical plant and flowers of ‘CATHARINA’ grown
outdoors in Abbenes, the Netherlands. The plant has been
growing in the ground for approximately 2 years. The photo-
graph was taken using conventional techniques and although
colors may appear different from actual colors due to light
reflectance it is as accurate as possible by conventional pho-
tographic techniques.

DETAILED BOTANICAL DESCRIPTION

In the following description, color references are made to
The Royal Horticultural Society Colour Chart except where
general terms of ordinary dictionary significance are used.
The following observations and measurements describe

‘CATHARINA’ plants grown outdoors in Abbennes, The Netherlands. Summer growing temperature ranged from approximately 14 to 33° C. during the day and 6 to 18° C. at night. Measurements and numerical values represent averages of typical flowering types. No chemical or photoperiodic treatments were given.

Botanical classification: *Echinacea purpurea* ‘CATHARINA.’

PROPAGATION

Time to initiate rooting: 7 to 10 days at approximately 25° C.
Time to produce a young rooted liner: Approximately 40 days at approximately 22° C.

Root description: Fine, well-branched, fibrous, becoming slightly woody with age.

PLANT

Growth habit: Hardy, upright, herbaceous perennial.

Height: Approximately 58.4 cm.

Plant spread: Approximately 28.2 cm.

Growth rate: Moderate. Approximately 15 cm per months during Spring months, while actively growing.

Branching characteristics: Basal branching, moderately free branching.

Length of lateral branches: Approximately 23.8 cm, excluding inflorescence.

Diameter of lateral branches: Approximately 0.4 cm.

Quantity of lateral branches: Approximately 7.

Texture of lateral branches: Densely covered with short, strigose hairs. Average hair length approximately 0.05 cm.
Hair color: Near RHS Green-White 157A.

Color of lateral branches: Near RHS Yellow-Green 146A.

Aspect: Approximately 25 angle from base of plant.

Strength of lateral branches: Very strong.

Internode length: Approximately 7.2 cm.

Age of plant described: Approximately 2 years old.

FOLIAGE

Leaf:

Arrangement.—Alternate, simple.

Average length.—Approximately 9.2 cm.

Average width.—Approximately 4.6 cm.

Shape of blade.—Narrowly ovate, slightly carinate.

Apex.—Acute to acuminate.

Base.—Attenuate.

Attachment.—Stalked.

Margin.—Sparsely serrate.

Texture of top surface.—Rough with a slightly glossy appearance. Densely covered with short strigose hairs. Average hair length approximately 0.03 cm, colored near RHS Green-White 157D.

Texture of bottom surface.—Rough with a slightly glossy appearance. Densely covered with short strigose hairs. Average hair length approximately 0.03 cm, colored near RHS Green-White 157D.

Leaf internode length.—Approximately 1.1 cm.

Color.—Young foliage upper side: Near RHS Green N137A. Young foliage under side: Near RHS Green 137B. Mature foliage upper side: Near RHS Green N137D. Mature foliage under side: Near RHS Green 137B.

Venation.—Type: Pinnate. Venation color upper side: Near RHS Green 138C. Venation color under side: Near RHS Yellow-Green 144D.

Petiole:

Length.—Average 3.9 cm.

Diameter.—Approximately 0.3 cm.

Texture.—Glossy, smooth.

Color.—Near RHS Green 138B.

FLOWER

Bloom period: Naturally blooming continuously from Summer to mid-Fall. During this time, blooming begins approximately 20 weeks after planting a rooted liner.

Bud:

Bud shape.—Flattened globular.

Bud length.—Approximately 1.8 cm.

Bud diameter.—Approximately 2.8 cm.

Bud color.—Near Yellow-Green 145B. Center near Green 143A.

Inflorescence:

Form.—Terminal capitulum consisting of numerous disc florets, one row of ray florets and three rows of involucre bracts.

Flower:

Diameter of entire flower.—Approximately 6.8 cm.

Depth of flower.—Approximately 3.1 cm.

Disc diameter.—Approximately 4.4 cm.

Receptacle shape.—Flattened globular.

Receptacle height.—Average 1.0 cm.

Receptacle diameter.—Average 1.1 cm.

Receptacle color.—Near RHS White N155B.

Quantity of inflorescences per lateral stem.—1.

Quantity of open inflorescences and buds per plant.—Average 7.

Longevity.—Average 5 weeks on the plant without significant fading or aging. Self-Cleaning.

Fragrance.—No fragrance.

Ray florets:

Arrangement.—Rotate.

Aspect/orientation.—Flat, approximately 120° angle from receptacle.

Shape.—Elliptic.

Number of ray florets.—Avg. 15.

Length.—Approximately 3.4 cm.

Width.—Approximately 1.1 cm.

Apex shape.—Emarginate.

Base.—Attenuate.

Margin.—Entire.

Texture.—Smooth, matte, moderately ribbed lengthwise.

Color.—Ray Florets Upper surface at first opening: Near RHS Red-Purple 70A. Upper surface at maturity: Near RHS Red-Purple 70B. Upper surface at fading: Not fading, color maintained. Under surface at first opening: Near RHS Red-Purple 61A, covered with a thin waxy appearing layer near Purple N77B. Under surface at maturity: Near RHS Red-Purple 70B covered with a thin waxy appearing layer near Purple N77B. Under surface at fading: Not fading, color maintained.

Disc florets:

Number of disc florets.—Approximately 500.

Arrangement.—Acropetally on disc.

Length.—Approximately 1.5 cm.

Width.—Approximately 0.4 cm.

Shape.—Tubular, lower $\frac{1}{3}$ fused, upper $\frac{2}{3}$ open.

Margin.—Entire.

Apex.—Acute.

Base.—Fused, acute.

Texture.—Smooth, glossy.

Color disc florets.—At first opening, only exterior visible: Near RHS Red-Purple 60B, base near Red-Purple 69C. At maturity, exterior surface: Near RHS Red-Purple 63C, tip near 63B base near Red-Purple 63D. At maturity, interior surface: Near RHS Red-Purple 61A.

Receptacle spines:

Number of spines.—Approximately 500.

Length.—Approximately 1 cm.

Width.—Approximately 0.1 cm.

Spine shape.—Acicular.

Spine apex.—Acute.

Spine base.—Attenuate.

Spine texture.—Smooth.

Spine apex color.—Near RHS Red-Purple 59A.

Spine mid-section color.—Near RHS Yellow-Green 144B.

Spine base color.—Near RHS Yellow-Green 145D.

Phyllaries/involucral bracts:

Arrangement.—Rotate, in 3 rows.

Number.—Approximately 75.

Length.—Approximately 0.9 cm.

Width.—Approximately 0.4 cm.

Shape.—Lanceolate.

Apex.—Acute.

Base.—Broadly cuneate.

Margin.—Entire.

Texture.—Sparsely pubescent near margins on both sides.

Color.—Inside color near RHS Green N137A. Outside color near RHS Green 137B.

PEDUNCLE

Length.—Average 15.8 cm.

Width.—Average 0.9 cm.

Texture.—Covered in strigose hairs.

Color.—Near RHS Yellow-Green 144A.

Strength.—Very strong.

REPRODUCTIVE ORGANS

Androecium:

Stamens not observed in any flowers to date.

Gynoecium: Present only disc florets.

Pistil length.—Approximately 0.9 cm.

Stigma shape.—Decurrent.

Stigma color.—Near RHS Greyed-Purple 187A.

Style length.—About 0.7 cm.

Style color.—Near RHS Green-White 157A.

Ovary color.—Near RHS Yellow-Green 145C.

OTHER CHARACTERISTICS

Disease resistance: Neither resistance nor susceptibility to diseases or pests normal to *Echinacea* have been observed in this variety.

Drought tolerance and temperature tolerance: Hardy perennial, low temperature tolerance to USDA Zone 4, at least. Tolerates high temperature extremely well, to at least 35°

C. Some drought tolerance. Plants can be allowed in dry soil for several days without damage.

Fruit/seed production: Fruits and seeds have not been observed.

What is claimed is:

1. A new and distinct cultivar of *Echinacea* plant named 'CATHARINA' as herein illustrated and described.

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Fig. 1